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CLASSIFICATION AND NEUTRAL NETWORKS

Random state	Classification agl1	Classification ag2	NN
0	0.54	0.54	0.98
1	0.53	0.53	1.0
2	0.54	0.54	0.96
3	0.55	0.55	0.96
4	0.54	0.54	0.19
5	0.53	0.53	0.98
6	0.54	0.54	0.98

MEAN	0.539	0.539	0.864
Highest	0.55	0.55	1.0
Lowest	0.53	0.53	0.19

Classification Models: Report with Model Modification

Introduction

Examining three machine learning models—Logistic Regression, Support Vector Machine (SVM), and Neural Networks (NN)—applied to a classification problem. This report presents the performance of each model, explains their working principles, and explores methods to enhance their performance through specific modifications. I have compared and analysed the result to their effectiveness.

Comparing Model Accuracies

The Table Above summarizes the accuracies of each model across multiple random states:

Observations:

- Both Logistic Regression and SVM demonstrate consistent but low accuracies (53-55%)from the mean .
- Neural Networks also outperform the 2 models with accuracies as high as 1.0 but exhibit variability, with a drastic drop to 0.19 at Random State 4.

Explanation of Model Functionality

Logistic Regression

- Logistic Regression is a linear model used for binary classification. It calculates the probability of a data point belonging to a specific class using the sigmoid function
- **Advantages**: Simple, interpretable, and computationally efficient.
- **Limitations**: Assumes a linear relationship between features and the target, which may not hold for complex datasets.

Reference to Mr Jerome Friedman . The Elements of Statistical Learning.(2009)

Support Vector Machine (SVM)

- SVMs find a hyperplane that maximizes the margin between data points of different classes. For non-linear problems, the kernel trick maps data to a higher-dimensional space.
- **Advantages**: Effective for high-dimensional spaces and works well with a clear margin of separation.
- **Limitations**: Computationally expensive for large datasets performance depends on kernel and hyperparameter selection.

Reference: Vladimir Vapnik Computer scientist Support vector network (in year 1995)

Neural Networks (NN)

Neural networks consist of layers of neurons. Each neuron computes a weighted sum of inputs, applies an activation function, and passes the result to the next layer. The model is trained by minimizing a loss function using backpropagation.

- **Advantages**: Capable of learning complex, non-linear relationships.
- **Limitations**: Prone to overfitting, requires more data, and is computationally intensive.

Reference: ian Goodfellow . (2016). Deep Learning.

Improving Models with Rationale and Implementation

Logistic Regression

Feature scaling and removal of irrelevant features can improve model performance by addressing multicollinearity and scaling disparities.

Implementation:

1. **Feature Selection**: Used correlation heatmaps to identify and drop features with low correlation to the target variable.
2. **Standardization**: Applied StandardScaler to ensure all features have zero mean and unit variance.

Support Vector Machine (SVM)

Rationale: SVM performance can be enhanced by tuning hyperparameters and using non-linear kernels like RBF (Radial Basis Function).

Implementation:

Neural Networks (NN)

Rationale: Overfitting and variability in NN performance can be addressed by adding dropout layers and reducing the number of epochs.

Implementation:

1. **Dropout Layers**: Added dropout after each hidden layer to prevent overfitting.
2. **Reduced Epochs**: Trained the model for 100 epochs with early stopping.

Conclusion

1. **Logistic Regression**: Achieved modest improvement with feature scaling and selection.
2. **SVM**: Hyperparameter tuning and non-linear kernels enhanced performance significantly.
3. **Neural Networks**: Dropout and early stopping reduced overfitting, resulting in more stable performance.

Other Future improvements that could include collecting more data, exploring additional features, and leveraging ensemble methods for better results.